

Rings will not be observable.

$$29 \quad \mathbf{A} \quad A_{P0} + A_{Q0} = 800 \quad (1)$$

$$A_{P0} \left(\frac{1}{2} \right)^{120/60} + A_{Q0} \left(\frac{1}{2} \right)^{120/30} = 80$$

$$\frac{1}{4} A_{P0} + \frac{1}{16} A_{Q0} = 80$$

$$\Rightarrow 4A_{P0} + A_{Q0} = 1280 \quad (2)$$

(2) – (1) :

$$3A_{P0} = 480$$

$$A_{P0} = 160$$

30. **B.** Only α -decay results in a change in mass number, and each α -decay causes the mass